

Lab 1 – PatientPath Descriptive Paper

Kaitlyn Molaison

Old Dominion University

CS411

Dr Sumaya Sanober

09/27/2026

Version 1

Table of Contents

1 Introduction.....	3
2 Product Description.....	7
2.1 Key Product Features and Capabilities.....	8
2.2 Major Components (Hardware/Software)	10
3 Identification of Case Study.....	11
4 Prototype Description.....	14
5 Glossary	17
6 References.....	18

List of Figures

Figure 1: Fax Flow.....	4
Figure 2: Impact.....	5

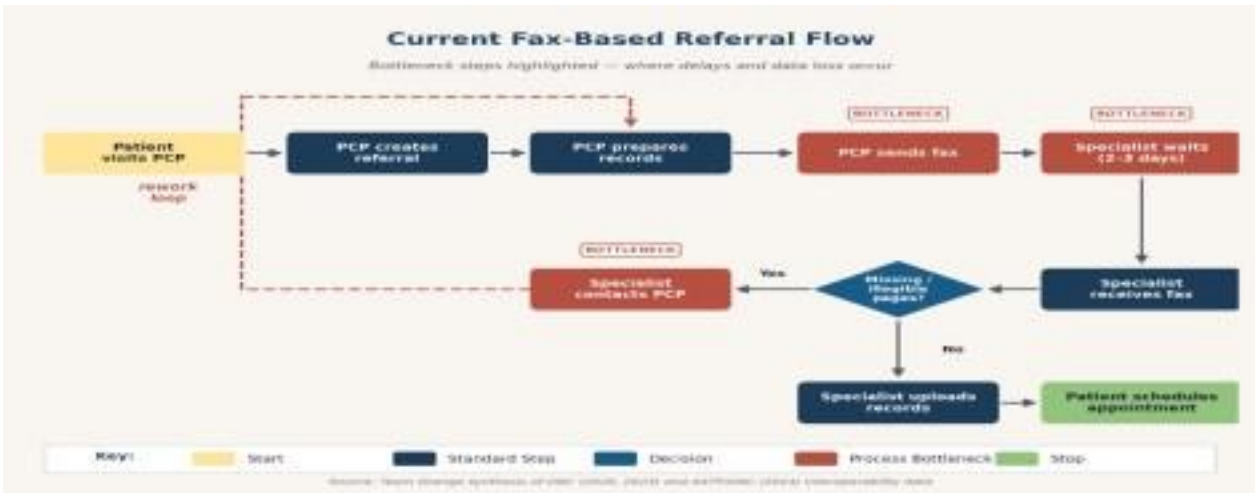
List of Tables

Table 1: Table of Comparison Between RWP and Prototype.....	16
---	----

1 Introduction

In the United States, healthcare continues to rely on outdated mechanisms for exchanging patient information. This creates measurable delays, administrative burden, and clinical risks. Despite the widespread availability of digital alternatives, approximately 65% of medical record transfers still occur by fax (ONC, 2023). Only 43% of hospitals routinely utilize all four core interoperability domains. These domains are sending, receiving, and integrating records (ASTP/ONC, 2024). The gap widens especially at the edges of the network. Only 16% of hospitals regularly share records with long term or post-acute providers that are outside their own network. 17% share records with behavioral health providers (ASTP/ONC, 2024). Among office-based physicians, 85% report difficulty transferring records across different EHR vendors. 71% work with referral partners who lack exchange capability at all (ONC, 2022). This results in a fragmented national system in which critical patient information is too often locked behind incompatible software, proprietary formats, and paper type workflows (see **Figure 1.**).

Figure 1

Fax Flow

These gaps translate into real harm for healthcare providers, their support staff, and their patients. Time sensitive referrals stall while staff resend, have to send multiple requests, and manually track status of paperwork (ONC, 2020). Handoffs between primary care and specialists are incomplete, and referral loops frequently fail to close (Patel et al., 2018). Missing information leads to unnecessary repeat diagnostics, duplicate imaging, and redundant orders for various lab work ups (De Swart et al., 2023). This drives up overall costs while delaying care. The burden falls hardest on rural and lower resourced organizations, which are the least likely to engage in routine interoperability (ASTP/ONC, 2024) and the most likely to be forced to operate on a restricted IT infrastructure and erratic connectivity (see **Figure 2**). A functional solution must therefore do more than just digitize existing workflows. It must be secure, interoperable with existing EHR systems, usable in real world network environments, and align with HIPAA requirements at every layer.

Figure 2

Impact

A meaningful solution in this space must address six interlocking problem characteristics. Current healthcare record exchange relies on inefficient transfer processes that can delay or compromise records, while disconnected systems limit interoperability across organizations. Outdated security protocols can also expose patient information to privacy risks, and poor usability can slow down adoption among healthcare professionals. In addition, rigid system architecture may struggle to scale as healthcare networks grow, while connectivity limitations can make online-only tools impractical for rural and lower-resourced environments. These challenges are intricately connected, meaning that addressing only one or two of them does not fully resolve the broader problem. For example, a platform that achieves interoperability but fails in low-bandwidth environments would still leave some of the populations that could benefit most from improved record exchange underserved.

PatientPath is our proposed solution. It provides a secure, centralized, web-based record sharing and referral coordination platform designed to bridge organizations and the existing EHR systems rather than replace them. Being built on an API first architecture with FHIR R4 and HL7 compatibility, PatientPath interweaves real time digital record exchange, automated referral tracking, role-based access control, end to end encryption, and offline synchronization for rural and low bandwidth clinics. This is specifically designed to target hospitals, private practices, specialists, rural providers, and patients as well. PatientPath is engineered to be HIPAA aligned by design. The remainder of this document details PatientPath's features, architecture, case study, and describes a functional prototype built to demonstrate the platform's core characteristics in a representative rural referral scenario.

2 Product Description

PatientPath is a secure, web-based platform that replaces the fax-driven exchange of medical records with real-time digital record sharing and automated referral coordination. Rather than competing with the EHR systems already in use, the platform sits between organizations as an interoperability layer: it ingests records from existing systems, normalizes them to FHIR R4, and routes them across organizational boundaries through a single API gateway. Providers, administrative staff, and patients each interact with PatientPath through a role-appropriate interface, while every transaction is encrypted, logged, and bound to explicit patient consent.

The goal of PatientPath is to eliminate the delays, document loss, and administrative burden that result from fax-based workflows, while strengthening the privacy and security posture that aging communication channels actively undermine. The platform's specific objectives are: (1) achieve cross-organization record exchange in real time using FHIR R4 and HL7 v2.x without requiring providers to replace their existing EHRs; (2) close the referral loop through a single form workflow with automated status tracking, stale-referral alerts, and supervisor escalation; (3) enforce HIPAA-aligned security through end-to-end encryption, role-based access control, multi factor authentication, and an immutable audit trail; (4) preserve operability in low-bandwidth or rural environments through local caching and queued synchronization on reconnection; and (5) sustain a 99.9% uptime target through cloud-hosted, horizontally scalable infrastructure.

2.1 Key Product Features and Capabilities

PatientPath delivers eight integrated capabilities that work together to replace fax-based workflows and address the problem characteristics identified in Section 1. User-facing capabilities include secure record upload, real-time record sharing, automated referral tracking, patient consent management, multi-factor authentication, and offline synchronization. Supporting infrastructure provides FHIR/HL7 interoperability, role-based access control, encryption, audit logging, and cloud-based availability.

Secure Record Upload — PatientPath validates uploaded files for format and completeness before encrypting and storing them in protected document storage. The production system is designed to complete typical uploads within five seconds under normal operating and network conditions.

Real-Time Record Sharing — When a referral is created, authorized receiving organizations are granted scoped, time-limited access to the appropriate patient information. Provider and specialist dashboards are updated automatically, eliminating the need for a manual record handoff.

Automated Referral Tracking — A single-form referral workflow supports automatic status transitions from Pending to In Review, Accepted, and Complete. Configurable inactivity thresholds can generate administrative alerts and escalate unresolved referrals to supervisors.

Patient Consent Management — Patients can grant or revoke access to their records through the patient portal. Revocation invalidates associated access permissions so that previously authorized sessions can no longer access the affected records.

Multi-Factor Authentication — TOTP-based multi-factor authentication provides an additional layer of account security. Repeated unsuccessful authentication attempts can trigger account lockout and administrative notification according to configurable security policies.

Offline Sync Capability — Recently accessed information can be cached locally when connectivity is unavailable. Changes made while offline are queued and automatically synchronized after connectivity is restored, with server-side data serving as the authoritative state during conflict resolution.

FHIR/HL7 Integration — A configurable translation layer maps EHR data to FHIR R4 and supports HL7 v2.x messaging, allowing information originating from different EHR systems to use a normalized representation for exchange.

End-to-End Encryption with RBAC — PatientPath protects stored information using AES-256 encryption and data in transit using TLS 1.3. Role-based access controls are enforced for API requests, while encryption keys are managed and rotated through a cloud key-management service according to the platform's security policy.

What makes PatientPath is significant is the combination of three properties that competing solutions do not offer together. First, the platform is EHR-agnostic and bridges existing systems rather than replacing them, meaning organizations adopt PatientPath without a multi-million-dollar EHR migration. Second, offline sync resilience makes the platform usable in rural and low-bandwidth clinics, which traditional digital fax replacements and cloud-only HIE solutions cannot serve. Third, the single-form referral workflow with immediate consent-to-access tokenization collapses what is currently a multi-step manual process into a single transaction, directly eliminating the bottlenecks responsible for the failure of fax-based transfers in the first place.

2.2 Major Components (Hardware/Software)

PatientPath does not require dedicated hardware on the customer side. The platform is hosted on commodity cloud infrastructure (AWS, Azure, or GCP, selected per deployment), and End users access the system through standard web browsers on any current desktop, tablet, or mobile device. Production deployment requires a load-balanced application tier behind a cloud managed API gateway, an active-passive PostgreSQL cluster with replication across availability zones for high availability, encrypted object storage for document blobs, and a managed key management service (AWS KMS or Azure Key Vault) for AES-256 key custody and rotation. Cached reads and queued writes for offline mode are held client-side in the browser through Service Worker storage and IndexedDB, requiring no installed software.

The software is structured according to the three-layer Major Functional Component Diagram (MFCD) defined in the team's design deck. The Presentation Layer contains four user-facing interfaces — Provider Dashboard, Patient Portal, Admin Console, and Login/MFA screens — along with the notification interface, all rendered as a React single-page application. The Application Layer sits behind a single API gateway and contains the platform's business logic, organized into four services: the Secure Record Exchange Service, Identity and Access Management, Patient Record Management, and the FHIR/HL7 Integration Service. The Data Layer holds encrypted document storage for medical records, a PostgreSQL database for users

and organizations, a separate FHIR-normalized data store, an append-only audit log database, and the consent records store.

The software stack consists of TypeScript with React on the front end and TypeScript with Node.js and Express on the back end. PostgreSQL serves as the primary relational database, accessed through Prisma as the ORM. WebSockets push real-time status updates to dashboards. Authentication is built on Speakeasy for TOTP-based MFA and OAuth 2.0 / OpenID Connect for hospital SSO integration. Encryption uses TLS 1.3 in transit and AES-256 at rest, with keys held in AWS KMS or Azure Key Vault. CI/CD runs through GitHub Actions, with Jest as the testing framework, and source control is managed in GitHub. The interoperability layer is built against the HL7 FHIR R4 specification with HL7 v2.x message support for legacy EHR queues.

3 Identification of Case Study

PatientPath is being developed for the small-to-mid-size primary care practices, specialty clinics, and federally qualified health centers (FQHCs) that experience the highest cost from fragmented record exchange and the lowest level of meaningful interoperability. As established in Section 1, only 16% of hospitals routinely share records with long-term and post-acute providers outside their own networks, and 71% of office-based physicians work with referral partners who lack any cross-organization exchange capability (ASTP/ONC, 2024; ONC, 2022). These figures concentrate on rural and lower-resourced settings where IT infrastructure is constrained and connectivity is unreliable — the exact populations for whom every additional fax round-trip translates into delayed care. PatientPath targets this segment because it is the segment that benefits most from a bridge-style interoperability platform that does not require an EHR replacement and remains usable when network connectivity is intermittent.

The case study group for prototype evaluation will consist of a representative cross-section of the four primary user roles: one to two primary care providers initiating referrals, one to two specialists receiving them, one administrative staff member responsible for referral coordination and record validation, one system administrator with responsibility for user provisioning and audit review, and two to three patient participants exercising the consent and transfer-visibility workflows from the patient portal. This group will be drawn from a single rural community health network so that the prototype can be evaluated end-to-end on a complete referral loop —

From the patient encounter through consent, transmission, receipt, and acknowledgment — within one organizational boundary the team can directly observe an instrument. Once validated against this initial group, PatientPath is positioned for expansion to a broader user base that shares the same structural pain points. Future users include behavioral health providers (only 17% of which routinely share records today), long-term and post-acute care facilities, multi-specialty group practices, regional Health Information Exchanges (HIEs) seeking a low-friction onboarding option for smaller member organizations, and larger hospital systems that need to coordinate referrals with the rural and specialty practices in their catchment area. The plugin-based EHR integration model is intended to support this expansion without requiring core platform changes for each new participating organization.

4 Prototype Description

The PatientPath prototype is a scoped implementation built to demonstrate the platform’s core value proposition — secure, real-time record sharing and closed-loop referral coordination — within a controlled environment that the case study group can exercise end-to-end. The prototype implements the same three-layer architecture as the end product (Presentation, Application, Data) and uses the same software stack (TypeScript, React, Node.js, PostgreSQL, Prisma) so that lessons from prototype evaluation transfer directly to the production building. What the prototype omits is the breadth, redundancy, and external integration surface that production deployment requires; it does not omit the security, consent, or audit functions, which are present in the prototype because removing them would invalidate the workflows being evaluated.

The prototype implements the seven user-facing workflows that anchor the design: secure record upload with AES-256 encryption, single-form referral creation with automatic specialist notification, real-time referral status tracking with stale-referral alerts, patient consent grant and revocation, multi-factor authentication, offline cache and sync queue for record reads, and the immutable audit log viewable from the admin console. All four user roles — primary care providers, specialists, administrative staff, and patients are functional, with role-based access control enforced at the API gateway. Records are exchanged in FHIR R4 normalized form within a sandbox EHR data set that simulates the variety of source-system formats the production platform must handle.

The prototype differs from the end product along five specific dimensions. First, EHR integration uses simulated EHR data rather than live adapters to Epic, Oracle, or eClinicalWorks; the production EHR adapter layer requires per-vendor authentication, sandbox certification, and customer-specific configuration that is out of scope for prototype evaluation. Second, the prototype runs in a single-region cloud deployment without active-passive failover or multi-AZ database replication; production operates against a 99.9% uptime SLA that requires the full HA configuration. Third, external services for key management, identity provider integration, and notification delivery use sandbox or development-tier services rather than production-grade KMS, hospital SSO, and SMS/email gateways. Fourth, the prototype is provisioned only for the case study group, while production supports unbounded onboarding through self-service organization registration. Fifth, the prototype demonstrates HIPAA-aligned controls but has not undergone formal third-party HIPAA audit, SOC 2 Type II certification, or external penetration testing, all of which are required before any handling of real PHI in production. Table 1 summarizes these differences.

Table 1*Comparison Between Real-World Product (RWP) and Prototype*

Dimension	Real-World Product	Prototype
EHR Integration	Live adapters for Epic, Oracle, and eClinicalWorks via FHIR R4 / HL7 v2.x	Simulated EHR data set in FHIR R4 format; no live vendor APIs
Deployment	Multi-region, multi-AZ, active passive failover. 99.9% uptime SLA	Single-region cloud deployment; no failover
Key Management	Production AWS KMS / Azure Key Vault with 90-day rotation	Sandbox KMS instance; manual rotation
Identity Provider	Hospital SSO via OAuth 2.0 / OpenID Connect	Local credential store with TOTP MFA
User Population	Unbounded; self-service organization onboarding	Case study group only (~7 users)

Notifications	Production SMS and email gateways	Logged in-app notifications only
Compliance	HIPAA audit, SOC 2 Type II, annual penetration testing	HIPAA-aligned controls implemented; no formal certification
Scale	Horizontal autoscaling under load	Fixed development-tier capacity

Offline Sync	Full reconciliation with server-side conflict resolution at scale	Demonstrable cache + queue against single test database
Audit Log	Append-only, exportable as CSV/PDF, long-term. retention	Append-only, viewable in admin console; short retention

5 Glossary

- **EHR:** Electronic Health Record
- **HIPAA:** Health Insurance Portability and Accountability Act
- **FHIR:** Fast Healthcare Interoperability Resources (HL7 standard for health data exchange)
- **HL7:** Health Level Seven (International healthcare data standards organization)

RBAC: Role-Based Access Control

- **MFA:** Multi-Factor Authentication
- **API:** Application Programming Interface
- **REST:** Representational State Transfer (architectural style for APIs)
- **HIE:** Health Information Exchange
- **PCP:** Primary Care Provider
- **FQHC:** Federally Qualified Health Center
- **CI/CD:** Continuous Integration / Continuous Deployment

LAB 1 – PATIENTPATH
DESCRIPTION 17

6 References

- Office of the National Coordinator for Health Information Technology. (2023, January). *Interoperability and methods of exchange among hospitals in 2021* (Data Brief No. 64). <https://healthit.gov/data/data-briefs/interoperability-and-methods-exchange-among-hospitals-2021/>
- Assistant Secretary for Technology Policy/Office of the National Coordinator for Health Information Technology. (2024). *Interoperable exchange of patient health information among U.S. hospitals: 2023* (Data Brief No. 71). <https://healthit.gov/data/data-briefs/interoperable-exchange-patient-health-information-among-us-hospitals-2023/>
- Office of the National Coordinator for Health Information Technology. (2022, July). *Interoperability among office-based physicians in 2019* (Data Brief No. 59). <https://healthit.gov/data/data-briefs/interoperability-among-office-based-physicians-2019/>
- Office of the National Coordinator for Health Information Technology. (2020, February). *Strategy on reducing regulatory and administrative burden relating to the use of health IT and EHRs*. <https://healthit.gov/wp-content/uploads/2020/02/BurdenReport.pdf>
- Patel, M. P., Schettini, P., O'Leary, C. P., Bosworth, H. B., Anderson, J. B., & Shah, K. P. (2018). Closing the referral loop: An analysis of primary care referrals to specialists in a large health system. *Journal of General Internal Medicine*, 33(5), 715–721. <https://doi.org/10.1007/s11606-018-4392-z>
- De Swart, M. E., Zonderhuis, B. M., Hellingman, T., Kuiper, B. I., Dickhoff, C., Heineman, D. J., Hendrickx, J. J., Kouwenhoven, M. C. M., Van Moorselaar, R. J. A., Schuur, M., Tenhagen, M., Van Der Velde, S., De Witt Hamer, P. C., Zijlstra, J. M., & Kazemier, G. (2023). Incomplete patient information exchange and unnecessary repeat diagnostics during oncological referrals in the Netherlands: Exploring the role of information exchange. *Health Informatics Journal*, 29(1), 14604582231153795. <https://doi.org/10.1177/14604582231153795>